TS: 2024 - 25 Board: TS
Class: X

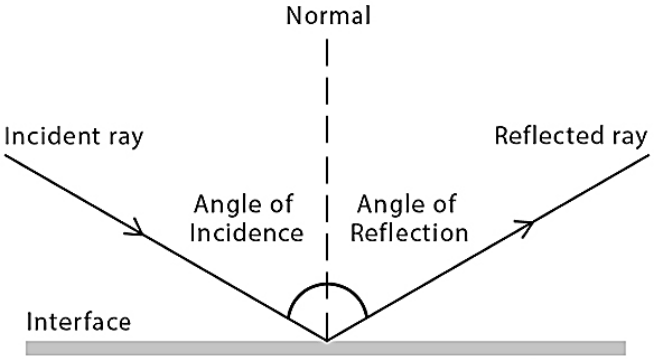
PHYSICS

By: Gayatri Patel

Reflection of Light at Curved Surfaces

Reflection:

The phenomenon of bouncing back of light ray in the same medium when incident on a smooth polished surface.



Incident ray:

The ray which is incident of a surface is called incident ray.

Reflected Ray:

The ray which is reflected from the surface is called a reflected ray.

Object distance (u):

The distance between object and mirror is called object distance.

Image distance (v):

The distance between image and mirror is called image distance.

Real image:

The image formed due to the converging of light rays is a real image.

Virtual image:

Virtual image ¡s the ¡mage formed due to diverging of light rays. (or) The Image formed due to the extension of rays in the backward direction

Curved Surface:

The surfaces which are not flat are called curved surface

The angle of incidence (∠i):

The angle between the incident ray and normal is called the angle of incidence.

The angle of reflection (∠r):

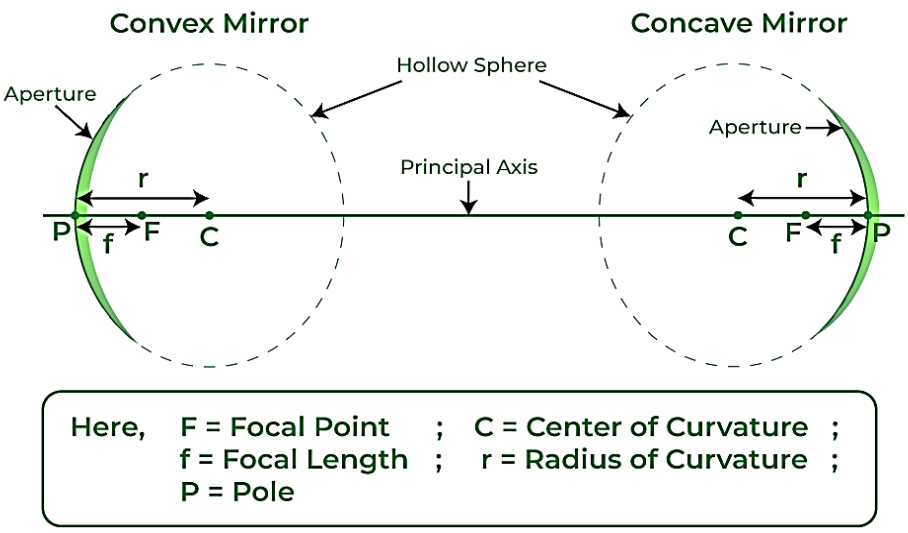
The angle between normal and reflected rays is called the angle of reflection.

Normal:

The Pine perpendicular to the surface is called normal.

Centre of curvature (C):

The normal from a concave mirror converges at a point that is called the center of curvature.



Spherical mirror:

A mirror that has a curved reflective surface is a spherical mirror.

Concave mirror:

- A spherical mirror whose reflecting surface is curved inwards is called a concave mirror
- The concave mirror converges the parallel light beam at one point. So it is called a converging mirror.

Convex mirror:

- A spherical mirror whose reflecting surface is curved outwards is called a convex mirror.
- Convex mirror diverges the parallel light beam. So it is called a diverging mirror.
- We always get a virtual and diminished image due to a convex mirror.

Aperture:

The part of the spherical mirror that is exposed to all the light rays that incident on it is called the aperture of the spherical mirror.

Centre of Curvature:

- The centre of hollow sphere of which the curved or spherical mirror forms a part is called the Centre of curvature(C).
- In Convex mirror, it lies behind the reflecting surface and in concave mirror it lies infront the reflecting surface.

Radius of curvature:

The radius of a hollow sphere of which the spherical mirror forms a part is called the radius of curvature. It is denoted by R.

Pole:

The center of the spherical surface is called its **pole**. It is denoted by *P*.

Principal Axis:

The line joining the center of curvature (C) and pole (P) of a spherical mirror and extended on either side is called the **principal axis**.

Principal Focus:

A point on the principal axis of a spherical mirror where the rays of light parallel to the principal axis meet or appear to meet after reflection from the spherical mirror is called the principal focus. It is denoted by *F*.

Focal Plane:

A plane normal or perpendicular to the principal axis and passing through the principal focus (F) of a spherical mirror is called the *focal plane* of the spherical mirror.

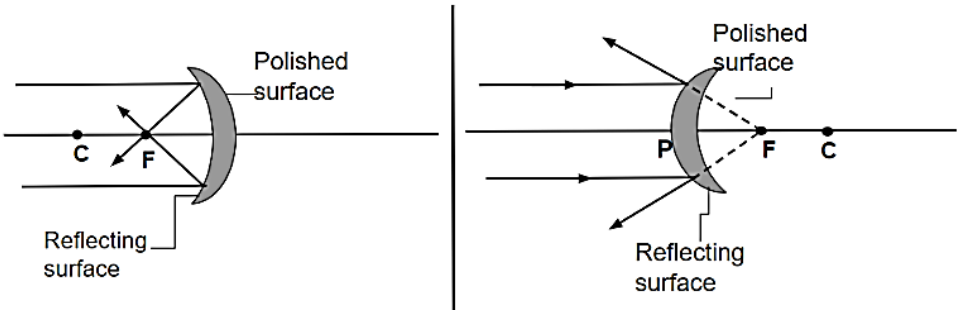
Focal Length:

The distance between the pole (P) and principal focus (f) of a spherical mirror is called the focal length of the mirror. It is denoted by f. As shown in the figure above, the focal length of the mirror is represented by PF.

Rules to draw Ray diagrams

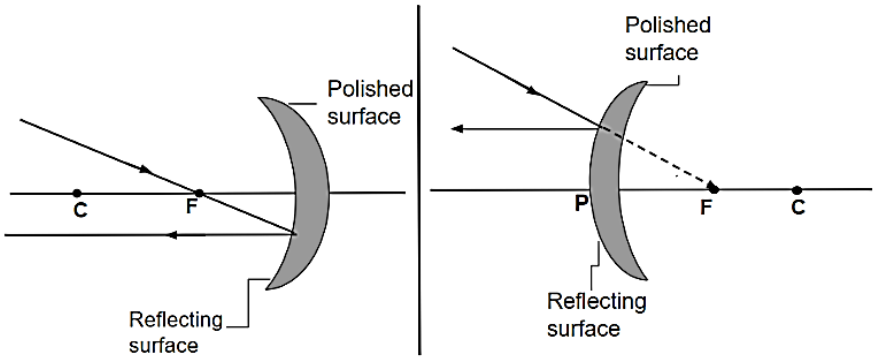
Rule-1:

- **For concave mirror:** A ray of light parallel to the principal axis will pass through the focus after reflection.
- **For Convex mirror:** A ray of light parallel to the principal axis will appear to come from focus.



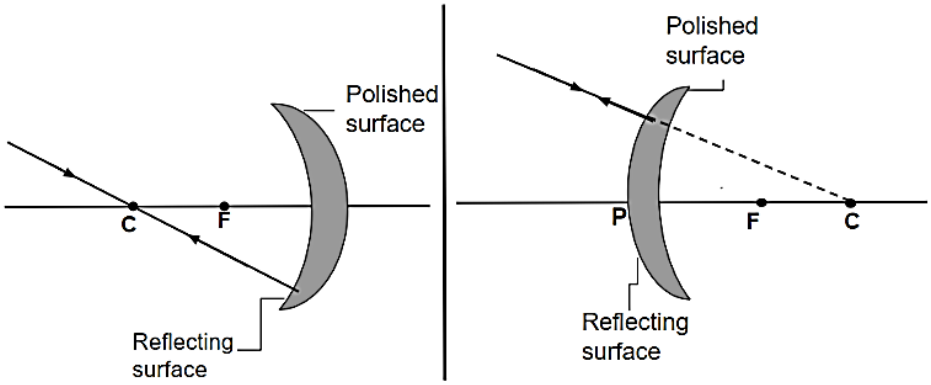
Rule 2:

- **For concave mirror:** A ray passing through the principal focus(F) becomes parallel to the principal axis.
- **For Convex mirror:** A ray directed towards principal focus becomes parallel to the principal axis.



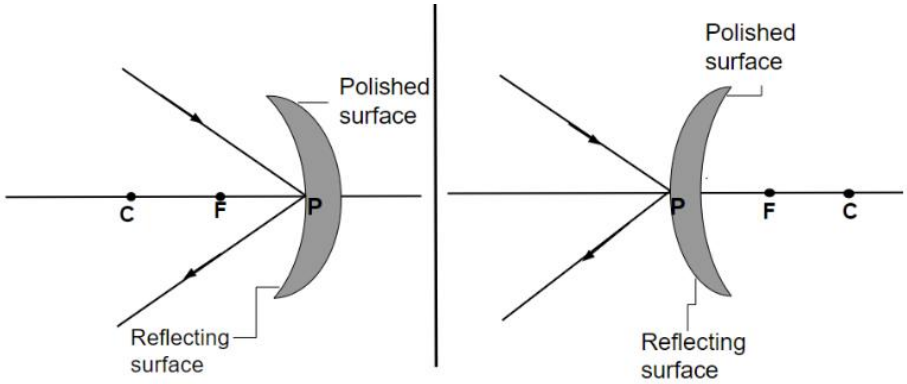
Rule 3:

- **For concave mirror:** A ray passing through the centre of curvature will take the same path after reflection.
- **For Convex mirror:** A ray directed towards the centre of curvature will take the same path after reflection.

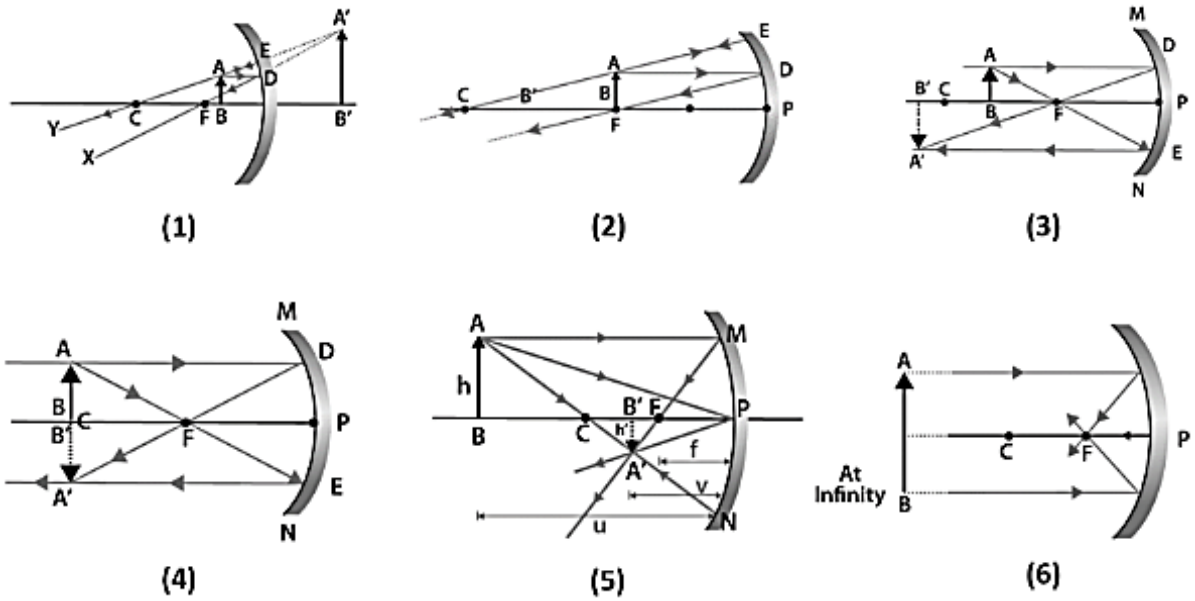


Rule 4:

- For any ray incident at any angel at the pole (P), the reflected ray follows the law of reflection for both concave as well as convex mirrors.



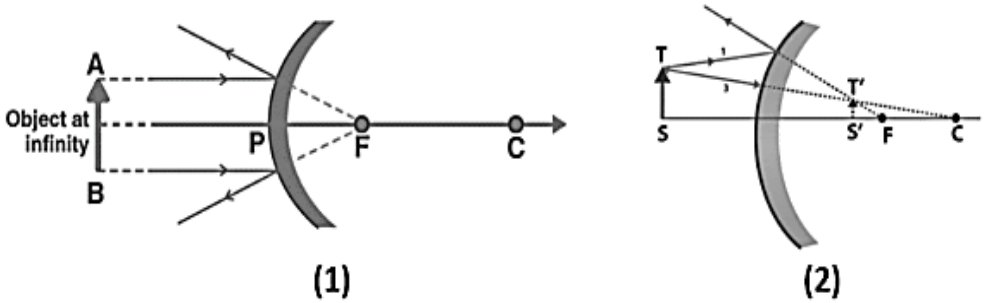
- **Image formation by Concave mirror.**



Position of the object	Position of the image	Enlarged?/ Diminished	Erected/ Inverted	Real / Virtual
Between mirror & F	Behind the mirror	Enlarged	Erected	Virtual
On focal point	At infinity	High enlarged	Inverted	Real
Between F and C	Beyond C	Enlarged	Inverted	Real
On centre of curvature	On C	Same size	Inverted	Real
Beyond C	Between F and C	Diminished	Inverted	Real

- **Image formation by Convex mirror**

S. No.	Position of the object	Position of the image	Image size	Nature of the image
1	Between infinity and the pole	Between pole and focus behind the mirror	Diminished	Virtual and erect
2	Infinity	At focus behind the mirror	Point size, Highly Diminished	Virtual and erect



Mirror Formula:

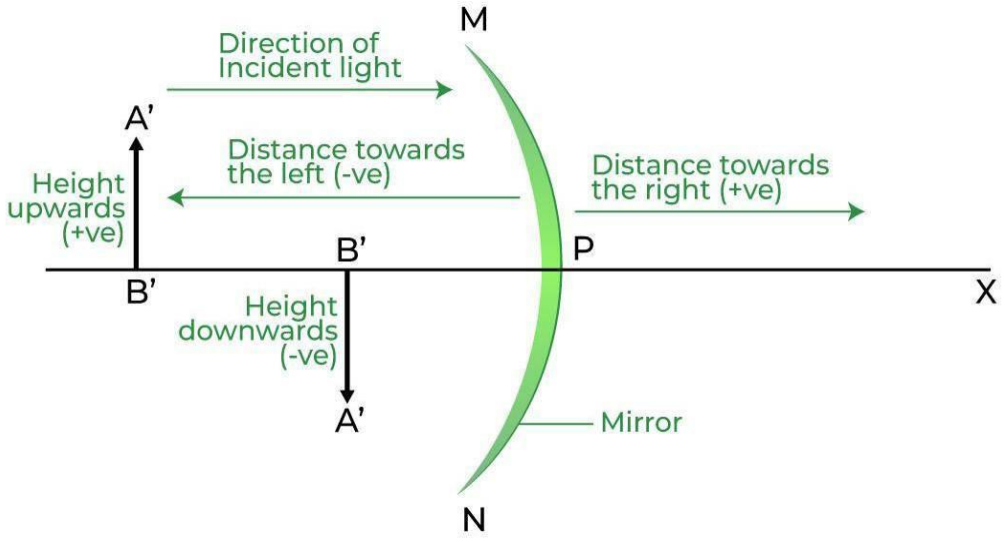
- The relation between object distance, image distance and Focal length is known as Mirror Formula.
- mirror formula $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

  f = focal length; u = object distance; v = image distance

Magnification:

- Linear magnification produced by a mirror is defined as the ratio of the size/height of the image to the size/height of the object. It has no units.
- magnification: $m = \frac{\text{size of the image}}{\text{size of the object}}$ (or) $\frac{h_1}{h_0}$ (or) $m = -\frac{\text{image distance}}{\text{object distance}} = -\frac{v}{u}$
- If "m" is positive - image is Virtual and erect
- "m" is negative - image is Real and Inverted
- If m=1, Size of the image is equal to size of the object
- If m>1, Image is Enlarged
- If m<1, image is diminished

Sign Convention for Spherical mirrors:

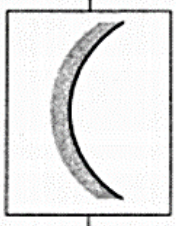


NOTE:

- According to the sign convention, the distances towards the left of the mirror are negative. Since an object is always placed to the left side of a mirror, therefore, the object distance (u) is always negative.
- The images formed by a concave mirror can be either behind the mirror (virtual) or in front of the mirror (real). So, the image distance (v) for a concave mirror can be either positive or negative depending on the position of the image.
- If the image is formed behind a concave mirror, the image distance (v) is positive but if the image is formed in front of the mirror, then the image distance will be negative.
- In a convex mirror, the image is always formed on the right-hand side (behind the mirror), so the image distance (o) for a convex mirror will always be positive.
- The focus of a concave mirror is in front of the mirror on the left side, so the focal length of a concave mirror will be negative (and written with a minus sign, say, -10 cm).
- On the other hand, the focus of the convex mirror is behind the mirror on the right side, so the focal length (and written with a plus sign, say +20 cm or just 20 cm), of a convex mirror is positive.
- The Focal Length and radius of curvature of a concave mirror are taken negatively.

LIGHT REFLECTION AT CURVED SURFACES

Concave mirror

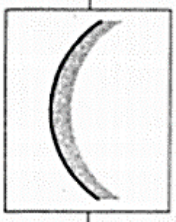


Converges parallel light rays at focus

Uses
Dentists
Car head lights
Solar cookers
Torch lights

It forms real, virtual images, enlarged, same size, diminished images, erected, inverted images

Convex mirror



Diverges light rays

Uses
Rear view mirrors
Stret view mirrors

It forms virtual, diminished images

Mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

v = image distance
u = object distance
f = focal length

Magnification formula

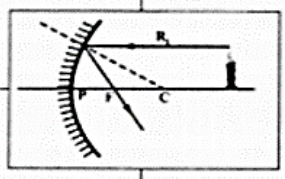
$$m = -\frac{v}{u}$$

$$m = \frac{h_i}{h_o}$$

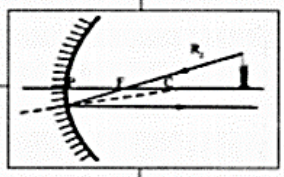
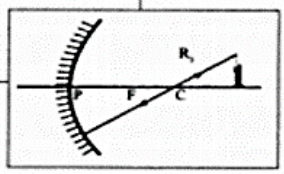
Ray diagrams

R_1 R_2 R_3 Rays

R_1



R_2



R_3